

Abu Dhabi Company for Onshore Petroleum Operations Ltd(ADCO)

SHAH

SY-120

SY-120

OH

Design: OH

Standard Survey Report

14 August, 2018

Survey Report

Company:	Abu Dhabi Company for Onshore Petroleum Operations Ltd(ADCO)	Local Co-ordinate Reference:	Well SY-120
Project:	SHAH	TVD Reference:	RTKB @ 552.0ft
Site:	SY-120	MD Reference:	RTKB @ 552.0ft
Well:	SY-120	North Reference:	Grid
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	EDMPRODR5K

Project	SHAH, SHAH		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	Nahrwan - U. Arab Emer.		
Map Zone:	Zone 39N (48 E to 54 E)		

Site	SY-120			
Site Position:		Northing:	2,537,145.00 m	Latitude: 22° 55' 8.489 N
From: Map		Easting:	790,712.00 m	Longitude: 53° 50' 2.951 E
Position Uncertainty:	0.0 ft	Slot Radius:	13.200 in	Grid Convergence: 1.10 °

Well	SY-120			
Well Position	+N/-S	0.0 ft	Northing:	2,537,145.00 m
	+E/-W	0.0 ft	Easting:	790,712.00 m
Position Uncertainty		0.0 ft	Wellhead Elevation:	ft
			Ground Level:	522.0 ft

Wellbore	OH				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	WMM_2005	31/12/2009	1.55	35.02	42,635.82694805

Design	OH			
Audit Notes:				
Version:	Phase:	ACTUAL	Tie On Depth:	0.0
Vertical Section:	Depth From (TVD) (ft)	+N/-S (ft)	+E/-W (ft)	Direction (°)
	242.0	0.0	0.0	39.87

Survey Program	Date	14/08/2018		
From (ft)	To (ft)	Survey (Wellbore)	Tool Name	Description
100.0	4,915.0	9 5/8" - 7" Gyro Survey (OH)	CT_GYRO_MS	Continuous Gyro Multishot
5,036.4	8,000.0	SY-120 6" OH Survey (OH)	MWD ISCWSA	MWD - Standard

Survey										
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)	
0.0	0.00	0.00	0.0	0.0	0.0	0.0	0.00	0.00	0.00	
100.0	0.12	149.44	100.0	-0.1	0.1	0.0	0.12	0.12	0.00	
125.0	0.15	165.76	125.0	-0.1	0.1	-0.1	0.19	0.12	65.28	
150.0	0.18	179.58	150.0	-0.2	0.1	-0.1	0.20	0.12	55.28	
175.0	0.21	193.40	175.0	-0.3	0.1	-0.2	0.22	0.12	55.28	
200.0	0.24	207.23	200.0	-0.4	0.0	-0.3	0.25	0.12	55.32	
225.0	0.35	235.49	225.0	-0.5	0.0	-0.4	0.72	0.44	113.04	
250.0	0.41	209.55	250.0	-0.6	-0.2	-0.6	0.72	0.24	-103.76	
275.0	0.35	182.37	275.0	-0.8	-0.2	-0.7	0.75	-0.24	-108.72	

Survey Report

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Site:	SY-120	MD Reference:	RTKB @ 552.0ft
Well:	SY-120	North Reference:	Grid
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	EDMPRODR5K

Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
300.0	0.22	173.26	300.0	-0.9	-0.2	-0.8	0.55	-0.52	-36.44
325.0	0.16	165.52	325.0	-1.0	-0.2	-0.9	0.26	-0.24	-30.96
350.0	0.15	166.68	350.0	-1.0	-0.2	-0.9	0.04	-0.04	4.64
375.0	0.14	184.56	375.0	-1.1	-0.2	-0.9	0.18	-0.04	71.52
400.0	0.10	190.06	400.0	-1.1	-0.2	-1.0	0.17	-0.16	22.00
425.0	0.07	185.01	425.0	-1.2	-0.2	-1.0	0.12	-0.12	-20.20
450.0	0.04	139.96	450.0	-1.2	-0.2	-1.0	0.20	-0.12	-180.20
475.0	0.06	156.05	475.0	-1.2	-0.2	-1.0	0.10	0.08	64.36
500.0	0.04	87.50	500.0	-1.2	-0.1	-1.0	0.23	-0.08	-274.20
525.0	0.04	97.30	525.0	-1.2	-0.1	-1.0	0.03	0.00	39.20
550.0	0.05	74.37	550.0	-1.2	-0.1	-1.0	0.08	0.04	-91.72
575.0	0.06	99.06	575.0	-1.2	-0.1	-1.0	0.10	0.04	98.76
600.0	0.05	75.75	600.0	-1.2	-0.1	-1.0	0.10	-0.04	-93.24
625.0	0.08	42.27	625.0	-1.2	0.0	-1.0	0.19	0.12	-133.92
650.0	0.04	55.72	650.0	-1.2	0.0	-0.9	0.17	-0.16	53.80
675.0	0.05	57.75	675.0	-1.2	0.0	-0.9	0.04	0.04	8.12
700.0	0.04	41.81	700.0	-1.2	0.0	-0.9	0.06	-0.04	-63.76
725.0	0.04	35.88	725.0	-1.2	0.0	-0.9	0.02	0.00	-23.72
750.0	0.07	38.44	750.0	-1.1	0.0	-0.9	0.12	0.12	10.24
775.0	0.08	41.36	775.0	-1.1	0.1	-0.8	0.04	0.04	11.68
800.0	0.10	46.54	800.0	-1.1	0.1	-0.8	0.09	0.08	20.72
825.0	0.08	30.81	825.0	-1.1	0.1	-0.7	0.13	-0.08	-62.92
850.0	0.09	52.46	850.0	-1.0	0.1	-0.7	0.13	0.04	86.60
875.0	0.16	55.09	875.0	-1.0	0.2	-0.7	0.28	0.28	10.52
900.0	0.13	38.91	900.0	-1.0	0.2	-0.6	0.20	-0.12	-64.72
925.0	0.13	37.29	925.0	-0.9	0.3	-0.5	0.01	0.00	-6.48
950.0	0.09	19.54	950.0	-0.9	0.3	-0.5	0.21	-0.16	-71.00
975.0	0.07	8.41	975.0	-0.8	0.3	-0.5	0.10	-0.08	-44.52
1,000.0	0.10	326.38	1,000.0	-0.8	0.3	-0.4	0.27	0.12	-168.12
1,025.0	0.05	289.73	1,025.0	-0.8	0.3	-0.4	0.27	-0.20	-146.60
1,050.0	0.03	322.82	1,050.0	-0.8	0.2	-0.4	0.12	-0.08	132.36
1,075.0	0.04	303.59	1,075.0	-0.8	0.2	-0.4	0.06	0.04	-76.92
1,100.0	0.05	272.32	1,100.0	-0.8	0.2	-0.4	0.10	0.04	-125.08
1,125.0	0.05	199.44	1,125.0	-0.8	0.2	-0.5	0.24	0.00	-291.52
1,150.0	0.06	169.44	1,150.0	-0.8	0.2	-0.5	0.12	0.04	-120.00
1,175.0	0.05	180.05	1,175.0	-0.8	0.2	-0.5	0.06	-0.04	42.44
1,200.0	0.08	155.02	1,200.0	-0.8	0.2	-0.5	0.16	0.12	-100.12
1,225.0	0.13	170.65	1,225.0	-0.9	0.2	-0.5	0.23	0.20	62.52
1,250.0	0.12	166.04	1,250.0	-0.9	0.2	-0.6	0.06	-0.04	-18.44
1,275.0	0.12	161.05	1,275.0	-1.0	0.2	-0.6	0.04	0.00	-19.96
1,300.0	0.15	165.54	1,300.0	-1.0	0.3	-0.6	0.13	0.12	17.96
1,325.0	0.12	153.55	1,325.0	-1.1	0.3	-0.7	0.16	-0.12	-47.96

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Well:	SY-120	North Reference:	Grid
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	EDMPRODR5K

Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
1,350.0	0.16	156.50	1,350.0	-1.2	0.3	-0.7	0.16	0.16	11.80
1,375.0	0.14	161.35	1,375.0	-1.2	0.3	-0.7	0.09	-0.08	19.40
1,400.0	0.11	148.06	1,400.0	-1.3	0.4	-0.7	0.17	-0.12	-53.16
1,425.0	0.08	154.70	1,425.0	-1.3	0.4	-0.8	0.13	-0.12	26.56
1,450.0	0.09	142.03	1,450.0	-1.3	0.4	-0.8	0.08	0.04	-50.68
1,475.0	0.11	145.55	1,475.0	-1.4	0.4	-0.8	0.08	0.08	14.08
1,500.0	0.09	146.25	1,500.0	-1.4	0.4	-0.8	0.08	-0.08	2.80
1,525.0	0.08	132.26	1,525.0	-1.4	0.5	-0.8	0.09	-0.04	-55.96
1,550.0	0.12	135.42	1,550.0	-1.5	0.5	-0.8	0.16	0.16	12.64
1,575.0	0.11	153.36	1,575.0	-1.5	0.5	-0.8	0.15	-0.04	71.76
1,600.0	0.07	126.79	1,600.0	-1.5	0.6	-0.8	0.23	-0.16	-106.28
1,625.0	0.06	116.16	1,625.0	-1.5	0.6	-0.8	0.06	-0.04	-42.52
1,650.0	0.03	313.16	1,650.0	-1.5	0.6	-0.8	0.36	-0.12	-652.00
1,675.0	0.03	38.44	1,675.0	-1.5	0.6	-0.8	0.16	0.00	341.12
1,700.0	0.06	309.03	1,700.0	-1.5	0.6	-0.8	0.27	0.12	-357.64
1,725.0	0.09	323.67	1,725.0	-1.5	0.6	-0.8	0.14	0.12	58.56
1,750.0	0.11	324.68	1,750.0	-1.5	0.5	-0.8	0.08	0.08	4.04
1,775.0	0.11	319.20	1,775.0	-1.4	0.5	-0.8	0.04	0.00	-21.92
1,800.0	0.11	318.78	1,800.0	-1.4	0.5	-0.8	0.00	0.00	-1.68
1,825.0	0.11	322.60	1,825.0	-1.4	0.4	-0.8	0.03	0.00	15.28
1,850.0	0.12	315.51	1,850.0	-1.3	0.4	-0.8	0.07	0.04	-28.36
1,875.0	0.12	315.98	1,875.0	-1.3	0.4	-0.7	0.00	0.00	1.88
1,900.0	0.14	310.41	1,900.0	-1.2	0.3	-0.7	0.09	0.08	-22.28
1,925.0	0.09	300.08	1,925.0	-1.2	0.3	-0.7	0.22	-0.20	-41.32
1,950.0	0.09	312.34	1,950.0	-1.2	0.3	-0.7	0.08	0.00	49.04
1,975.0	0.09	291.01	1,975.0	-1.2	0.2	-0.8	0.13	0.00	-85.32
2,000.0	0.09	296.31	2,000.0	-1.2	0.2	-0.8	0.03	0.00	21.20
2,025.0	0.08	278.00	2,025.0	-1.1	0.2	-0.8	0.12	-0.04	-73.24
2,050.0	0.06	251.54	2,050.0	-1.1	0.1	-0.8	0.15	-0.08	-105.84
2,075.0	0.11	90.98	2,075.0	-1.1	0.1	-0.8	0.67	0.20	-642.24
2,100.0	0.11	69.99	2,100.0	-1.1	0.2	-0.8	0.16	0.00	-83.96
2,125.0	0.14	79.03	2,125.0	-1.1	0.2	-0.7	0.14	0.12	36.16
2,150.0	0.18	72.39	2,150.0	-1.1	0.3	-0.7	0.18	0.16	-26.56
2,175.0	0.16	49.33	2,175.0	-1.1	0.4	-0.6	0.28	-0.08	-92.24
2,200.0	0.12	34.95	2,200.0	-1.0	0.4	-0.5	0.21	-0.16	-57.52
2,225.0	0.09	27.71	2,225.0	-1.0	0.4	-0.5	0.13	-0.12	-28.96
2,250.0	0.14	50.77	2,250.0	-1.0	0.5	-0.4	0.27	0.20	92.24
2,275.0	0.28	85.13	2,275.0	-0.9	0.5	-0.4	0.73	0.56	137.44
2,300.0	0.72	93.59	2,300.0	-0.9	0.8	-0.2	1.78	1.76	33.84
2,325.0	1.13	92.15	2,325.0	-1.0	1.2	0.0	1.64	1.64	-5.76
2,350.0	1.83	90.36	2,350.0	-1.0	1.8	0.4	2.81	2.80	-7.16
2,375.0	2.46	82.81	2,375.0	-0.9	2.7	1.1	2.76	2.52	-30.20
2,400.0	2.85	73.53	2,399.9	-0.7	3.9	2.0	2.32	1.56	-37.12

Survey Report

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Well:	SY-120	North Reference:	Grid
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	EDMPRODR5K

Survey										
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)	
2,425.0	3.04	61.12	2,424.9	-0.2	5.0	3.1	2.66	0.76	-49.64	
2,450.0	3.28	49.43	2,449.9	0.6	6.2	4.4	2.74	0.96	-46.76	
2,475.0	3.45	44.09	2,474.8	1.6	7.2	5.9	1.43	0.68	-21.36	
2,500.0	3.60	40.74	2,499.8	2.8	8.3	7.4	1.02	0.60	-13.40	
2,525.0	3.91	36.34	2,524.7	4.0	9.3	9.1	1.69	1.24	-17.60	
2,550.0	4.44	30.28	2,549.7	5.6	10.3	10.9	2.76	2.12	-24.24	
2,575.0	5.18	27.01	2,574.6	7.4	11.3	12.9	3.16	2.96	-13.08	
2,600.0	5.76	26.45	2,599.5	9.5	12.4	15.2	2.33	2.32	-2.24	
2,625.0	6.34	26.33	2,624.3	11.9	13.5	17.8	2.32	2.32	-0.48	
2,650.0	6.96	27.48	2,649.1	14.5	14.8	20.6	2.54	2.48	4.60	
2,675.0	7.78	29.80	2,673.9	17.3	16.4	23.8	3.49	3.28	9.28	
2,700.0	8.79	31.89	2,698.7	20.4	18.2	27.3	4.22	4.04	8.36	
2,725.0	9.79	33.38	2,723.4	23.8	20.4	31.3	4.11	4.00	5.96	
2,750.0	10.83	33.79	2,747.9	27.5	22.9	35.8	4.17	4.16	1.64	
2,775.0	11.68	34.10	2,772.5	31.6	25.6	40.6	3.41	3.40	1.24	
2,800.0	12.52	33.84	2,796.9	35.9	28.5	45.8	3.37	3.36	-1.04	
2,825.0	13.37	33.44	2,821.3	40.6	31.6	51.4	3.42	3.40	-1.60	
2,850.0	14.16	33.14	2,845.6	45.5	34.9	57.3	3.17	3.16	-1.20	
2,875.0	15.00	32.84	2,869.8	50.8	38.3	63.6	3.37	3.36	-1.20	
2,900.0	15.64	32.69	2,893.9	56.4	41.9	70.1	2.56	2.56	-0.60	
2,925.0	16.37	32.16	2,917.9	62.2	45.6	77.0	2.98	2.92	-2.12	
2,950.0	17.39	31.61	2,941.8	68.4	49.4	84.1	4.13	4.08	-2.20	
2,975.0	18.18	31.32	2,965.6	74.9	53.4	91.7	3.18	3.16	-1.16	
3,000.0	18.74	31.34	2,989.3	81.6	57.5	99.5	2.24	2.24	0.08	
3,025.0	19.22	31.34	3,013.0	88.6	61.8	107.6	1.92	1.92	0.00	
3,050.0	19.86	31.38	3,036.5	95.7	66.1	115.8	2.56	2.56	0.16	
3,075.0	20.46	31.74	3,060.0	103.1	70.6	124.4	2.45	2.40	1.44	
3,100.0	21.10	31.78	3,083.4	110.6	75.3	133.1	2.56	2.56	0.16	
3,125.0	21.86	31.79	3,106.6	118.4	80.1	142.2	3.04	3.04	0.04	
3,150.0	22.70	32.14	3,129.8	126.4	85.1	151.6	3.40	3.36	1.40	
3,175.0	23.76	32.48	3,152.7	134.8	90.4	161.4	4.27	4.24	1.36	
3,200.0	25.17	32.89	3,175.5	143.5	96.0	171.6	5.68	5.64	1.64	
3,225.0	26.23	33.00	3,198.0	152.6	101.9	182.4	4.24	4.24	0.44	
3,250.0	27.31	33.34	3,220.3	162.0	108.1	193.6	4.36	4.32	1.36	
3,275.0	28.56	33.62	3,242.4	171.8	114.5	205.2	5.03	5.00	1.12	
3,300.0	29.70	33.77	3,264.3	181.9	121.3	217.3	4.57	4.56	0.60	
3,325.0	30.68	33.98	3,285.9	192.3	128.3	229.8	3.94	3.92	0.84	
3,350.0	31.73	33.83	3,307.3	203.1	135.5	242.7	4.21	4.20	-0.60	
3,375.0	32.53	33.68	3,328.4	214.1	142.9	255.9	3.22	3.20	-0.60	
3,400.0	33.16	33.48	3,349.4	225.4	150.4	269.4	2.56	2.52	-0.80	
3,425.0	33.80	33.32	3,370.3	236.9	158.0	283.1	2.58	2.56	-0.64	
3,450.0	34.33	33.30	3,391.0	248.6	165.7	297.0	2.12	2.12	-0.08	

Survey Report

Company:	Abu Dhabi Company for Onshore Petroleum Operations Ltd(ADCO)	Local Co-ordinate Reference:	Well SY-120
Project:	SHAH	TVD Reference:	RTKB @ 552.0ft
Site:	SY-120	MD Reference:	RTKB @ 552.0ft
Well:	SY-120	North Reference:	Grid
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	EDMPRODR5K

Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
3,475.0	34.92	33.22	3,411.6	260.5	173.5	311.1	2.37	2.36	-0.32
3,500.0	35.55	32.94	3,432.0	272.6	181.3	325.5	2.60	2.52	-1.12
3,525.0	36.15	32.77	3,452.3	284.9	189.3	340.0	2.43	2.40	-0.68
3,550.0	37.11	32.42	3,472.3	297.5	197.3	354.8	3.93	3.84	-1.40
3,575.0	37.93	32.00	3,492.1	310.3	205.4	369.9	3.44	3.28	-1.68
3,600.0	38.41	32.00	3,511.8	323.4	213.6	385.2	1.92	1.92	0.00
3,625.0	38.88	31.90	3,531.3	336.7	221.9	400.6	1.90	1.88	-0.40
3,650.0	39.44	32.04	3,550.7	350.1	230.2	416.3	2.27	2.24	0.56
3,675.0	39.53	32.22	3,570.0	363.6	238.7	432.0	0.58	0.36	0.72
3,700.0	39.61	32.30	3,589.3	377.0	247.2	447.8	0.38	0.32	0.32
3,725.0	39.48	32.28	3,608.6	390.5	255.7	463.6	0.52	-0.52	-0.08
3,750.0	39.64	32.15	3,627.8	403.9	264.2	479.4	0.72	0.64	-0.52
3,775.0	39.59	32.02	3,647.1	417.5	272.6	495.2	0.39	-0.20	-0.52
3,800.0	40.13	31.98	3,666.3	431.0	281.1	511.0	2.16	2.16	-0.16
3,825.0	40.74	31.97	3,685.3	444.8	289.7	527.1	2.44	2.44	-0.04
3,850.0	41.75	32.11	3,704.1	458.8	298.5	543.4	4.06	4.04	0.56
3,875.0	42.69	32.63	3,722.6	473.0	307.5	560.1	4.01	3.76	2.08
3,900.0	43.53	33.29	3,740.9	487.3	316.8	577.0	3.81	3.36	2.64
3,925.0	44.61	33.61	3,758.8	501.8	326.3	594.3	4.41	4.32	1.28
3,950.0	45.07	33.90	3,776.6	516.5	336.1	611.9	2.01	1.84	1.16
3,975.0	45.61	33.97	3,794.1	531.2	346.1	629.5	2.17	2.16	0.28
4,000.0	46.45	33.78	3,811.5	546.1	356.1	647.4	3.40	3.36	-0.76
4,025.0	47.01	33.42	3,828.6	561.3	366.2	665.5	2.47	2.24	-1.44
4,050.0	47.22	33.25	3,845.6	576.6	376.2	683.7	0.98	0.84	-0.68
4,075.0	48.01	33.23	3,862.5	592.1	386.4	702.1	3.16	3.16	-0.08
4,100.0	49.19	33.36	3,879.0	607.7	396.6	720.7	4.74	4.72	0.52
4,125.0	50.15	33.44	3,895.2	623.6	407.1	739.6	3.85	3.84	0.32
4,150.0	50.49	33.38	3,911.2	639.7	417.7	758.8	1.37	1.36	-0.24
4,175.0	50.65	33.07	3,927.0	655.9	428.3	777.9	1.15	0.64	-1.24
4,200.0	50.94	32.35	3,942.8	672.2	438.8	797.2	2.52	1.16	-2.88
4,225.0	51.74	31.64	3,958.5	688.7	449.1	816.5	3.89	3.20	-2.84
4,250.0	52.68	31.65	3,973.8	705.5	459.5	836.0	3.76	3.76	0.04
4,275.0	54.16	31.86	3,988.7	722.6	470.1	855.9	5.96	5.92	0.84
4,300.0	56.32	32.02	4,002.9	740.0	480.9	876.3	8.66	8.64	0.64
4,325.0	57.98	32.44	4,016.5	757.8	492.1	897.1	6.79	6.64	1.68
4,350.0	58.42	32.65	4,029.7	775.7	503.6	918.2	1.90	1.76	0.84
4,375.0	59.57	33.07	4,042.5	793.7	515.2	939.4	4.82	4.60	1.68
4,400.0	61.28	33.73	4,054.9	811.9	527.1	961.0	7.22	6.84	2.64
4,425.0	62.67	33.84	4,066.6	830.2	539.4	983.0	5.57	5.56	0.44
4,450.0	64.49	33.66	4,077.8	848.8	551.9	1,005.2	7.31	7.28	-0.72
4,475.0	66.54	33.90	4,088.1	867.7	564.5	1,027.8	8.25	8.20	0.96
4,500.0	67.01	34.41	4,098.0	886.7	577.4	1,050.7	2.65	1.88	2.04
4,525.0	67.04	34.41	4,107.7	905.7	590.4	1,073.6	0.12	0.12	0.00

Survey Report

Company:	Abu Dhabi Company for Onshore Petroleum Operations Ltd(ADCO)	Local Co-ordinate Reference:	Well SY-120
Project:	SHAH	TVD Reference:	RTKB @ 552.0ft
Site:	SY-120	MD Reference:	RTKB @ 552.0ft
Well:	SY-120	North Reference:	Grid
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	EDMPRODR5K

Survey										
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)	
4,550.0	66.88	33.76	4,117.5	924.8	603.3	1,096.5	2.48	-0.64	-2.60	
4,575.0	67.04	32.73	4,127.3	944.0	615.9	1,119.4	3.84	0.64	-4.12	
4,600.0	66.71	31.50	4,137.1	963.5	628.1	1,142.1	4.71	-1.32	-4.92	
4,625.0	66.91	31.02	4,147.0	983.1	640.1	1,164.9	1.94	0.80	-1.92	
4,650.0	67.09	30.47	4,156.7	1,002.9	651.8	1,187.6	2.15	0.72	-2.20	
4,675.0	68.05	30.49	4,166.3	1,022.8	663.5	1,210.4	3.84	3.84	0.08	
4,700.0	69.55	31.18	4,175.3	1,042.8	675.5	1,233.4	6.53	6.00	2.76	
4,725.0	70.91	32.18	4,183.8	1,062.9	687.9	1,256.7	6.62	5.44	4.00	
4,750.0	72.23	33.29	4,191.7	1,082.8	700.7	1,280.2	6.75	5.28	4.44	
4,775.0	73.66	34.73	4,199.0	1,102.6	714.0	1,304.0	7.94	5.72	5.76	
4,800.0	74.70	36.33	4,205.8	1,122.2	728.0	1,328.0	7.43	4.16	6.40	
4,825.0	75.74	37.09	4,212.2	1,141.6	742.5	1,352.1	5.09	4.16	3.04	
4,850.0	77.37	37.53	4,218.0	1,160.9	757.2	1,376.4	6.74	6.52	1.76	
4,875.0	79.28	38.11	4,223.1	1,180.2	772.2	1,400.9	7.97	7.64	2.32	
4,900.0	81.31	38.75	4,227.3	1,199.5	787.5	1,425.5	8.50	8.12	2.56	
4,915.0	82.58	38.88	4,229.4	1,211.1	796.8	1,440.3	8.51	8.47	0.87	
5,036.4	89.49	38.61	4,237.8	1,305.5	872.6	1,561.4	5.70	5.69	-0.22	
5,130.0	89.80	39.27	4,238.4	1,378.3	931.4	1,654.9	0.78	0.33	0.71	
5,223.3	89.49	43.14	4,238.9	1,448.5	992.9	1,748.3	4.16	-0.33	4.14	
5,315.5	89.58	41.33	4,239.7	1,516.7	1,054.8	1,840.3	1.97	0.10	-1.96	
5,409.6	89.76	43.57	4,240.2	1,586.2	1,118.4	1,934.3	2.39	0.19	2.38	
5,502.8	89.73	45.82	4,240.6	1,652.5	1,183.9	2,027.2	2.41	-0.03	2.41	
5,596.0	89.73	44.99	4,241.1	1,717.8	1,250.2	2,119.9	0.89	0.00	-0.89	
5,687.4	89.70	43.28	4,241.5	1,783.5	1,313.9	2,211.1	1.87	-0.03	-1.87	
5,782.1	89.33	42.19	4,242.3	1,853.0	1,378.2	2,305.7	1.22	-0.39	-1.15	
5,875.0	89.86	39.96	4,243.0	1,923.0	1,439.2	2,398.5	2.47	0.57	-2.40	
5,968.4	89.70	42.87	4,243.3	1,993.1	1,501.0	2,491.9	3.12	-0.17	3.11	
6,154.8	89.55	33.90	4,244.6	2,139.0	1,616.6	2,678.0	4.81	-0.08	-4.81	
6,248.8	89.91	38.25	4,245.0	2,215.0	1,671.9	2,771.8	4.64	0.38	4.63	
6,342.2	89.70	39.49	4,245.3	2,287.7	1,730.6	2,865.2	1.35	-0.22	1.33	
6,435.9	89.64	42.51	4,245.9	2,358.4	1,792.0	2,958.9	3.22	-0.06	3.22	
6,528.2	89.76	45.81	4,246.4	2,424.6	1,856.3	3,050.9	3.58	0.13	3.58	
6,621.7	89.89	46.59	4,246.6	2,489.4	1,923.8	3,143.8	0.85	0.14	0.83	
6,714.9	89.86	46.65	4,246.8	2,553.3	1,991.5	3,236.3	0.07	-0.03	0.06	
6,808.5	89.76	48.53	4,247.2	2,616.5	2,060.6	3,329.1	2.01	-0.11	2.01	
6,902.6	89.80	48.66	4,247.5	2,678.7	2,131.2	3,422.1	0.14	0.04	0.14	
6,995.8	89.76	46.34	4,247.9	2,741.6	2,199.9	3,514.4	2.49	-0.04	-2.49	
7,089.0	89.76	46.43	4,248.3	2,806.0	2,267.4	3,607.1	0.10	0.00	0.10	
7,182.8	89.89	45.84	4,248.5	2,871.0	2,335.0	3,700.3	0.64	0.14	-0.63	
7,276.3	89.89	45.90	4,248.7	2,936.1	2,402.1	3,793.3	0.06	0.00	0.06	
7,369.3	88.90	41.78	4,249.7	3,003.1	2,466.5	3,886.1	4.56	-1.06	-4.43	
7,461.9	89.67	40.87	4,250.9	3,072.6	2,527.6	3,978.6	1.29	0.83	-0.98	

Survey Report

Company:	Abu Dhabi Company for Onshore Petroleum Operations Ltd(ADCO)	Local Co-ordinate Reference:	Well SY-120
Project:	SHAH	TVD Reference:	RTKB @ 552.0ft
Site:	SY-120	MD Reference:	RTKB @ 552.0ft
Well:	SY-120	North Reference:	Grid
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	EDMPRODR5K

Survey										
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)	
7,554.4	89.76	42.64	4,251.3	3,141.7	2,589.3	4,071.1	1.91	0.10	1.91	
7,648.5	89.70	43.99	4,251.8	3,210.1	2,653.8	4,165.0	1.44	-0.06	1.44	
7,741.7	89.67	43.55	4,252.3	3,277.4	2,718.3	4,258.0	0.47	-0.03	-0.47	
7,836.4	89.67	42.83	4,252.8	3,346.5	2,783.1	4,352.5	0.76	0.00	-0.76	
7,929.0	89.67	43.24	4,253.4	3,414.2	2,846.3	4,445.0	0.44	0.00	0.44	
7,972.6	89.64	43.16	4,253.6	3,445.9	2,876.1	4,488.5	0.20	-0.07	-0.18	
8,000.0	89.70	43.40	4,253.8	3,465.9	2,894.9	4,515.9	0.90	0.22	0.88	

Casing Points					
Measured Depth (ft)	Vertical Depth (ft)	Name	Casing Diameter (in)	Hole Diameter (in)	
210.0	210.0	CONDUCTOR	13.375	13.375	
4,176.0	3,927.7	PRODUCTION LINER	9.625	12.250	
4,997.0	4,236.7	PRODUCTION LINER	7.000	8.500	

Checked By: _____	Approved By: _____	Date: _____
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